

PROJECT OVERVIEW

PainEase PCA Advanced PP3500 program

A patient-controlled analgesia infusion pump — with connectivity adapter
and Cloud Suite
510(k) with Predetermined Change Control Plan (PCCP)

Device	PP3500 — PainEase PCA Advanced
Predicate	PP3000 / K190567 (illustrative)
Composition	SaMD + SiMD firmware + custom medical-electrical hardware
DHFs	10 total — 3 top-level + 7 nested under Cloud Suite
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AGENDA

How this deck is organized

Five content sections plus the appendix. ~20 minutes.

01 Project Overview & Strategy Device composition, DHF topology, regulatory approach (510(k) + PCCP).

02 Agentic Approach Project shape, structure, skills, and 23 persona agents.

03 Quality & Process How skills, hooks, and rules turn conventions into automatic guardrails.

04 Agents advise. Humans decide. Cross-functional advisory panel with the human sign-off always required.

05 The Project Console Local FastAPI browser app — agents, documents, dashboards, trace matrix.

PROJECT OVERVIEW & STRATEGY

What PP3500 is

Single composite system — SaMD + SiMD firmware + custom medical-electrical hardware.

PCA Device

pca-device · Class II medical device · IEC 62304 Class C

SiMD pump firmware + SaMD modules + custom hardware. On-pump delivery, dose-limit enforcement, alarms, touchscreen UI, barcode scan, self-test. The 510(k) anchor DHF.

Connectivity Adapter

connectivity-adapter · MDDS (non-device, post-2015 FDA)

On-prem bridge carrying PCA telemetry + drug-library pushes to and from the Cloud Suite. Named as adjacent infrastructure in the PP3500 510(k); only its cybersecurity posture is pulled in.

Cloud Suite

cloud-suite · platform DHF · 7 child DHFs

Drug Library Manager is Class II SaMD (PCA accessory, mutates the dose-limit table). The other six children (Fleet, Compliance Reports, Analytics, Inventory, Alerts, Clinical Interface) are non-device software.

PROJECT OVERVIEW & STRATEGY

Regulatory strategy

One 510(k) for the PCA device, with an integrated PCCP — pre-authorized post-market change.

Filing scope

- **PP3500 510(k)** covers only the PCA device itself.
- **Connectivity Adapter** MDDS — not separately filed, cybersecurity only.
- **Drug Library Manager** Class II SaMD accessory — bundled vs. standalone TBD.
- **Other Cloud Suite components** non-device — out of filing.
- **Predicate** PP3000 / K190567 (illustrative).

Ct* carve-out

- **CtS** Critical to Safety — failure can cause patient harm.
- **CtF** Critical to Function — failure breaks the therapy.
- **CtC** Critical to Compliance — required by standard.
- **CtP** Critical to Performance — degrades claimed spec.
- **In scope = any Ct* tag.** Commercial-only requirements ship post-clearance.

PCCP envelope

- **Drug-library updates** new drugs, limit changes inside bounds.
- **Firmware patches** against a fixed, frozen risk profile.
- **Predictive-alarm SaMDs** that meet change-protocol acceptance criteria.
- **Out of envelope** anything that changes the risk profile or indications.

PROJECT OVERVIEW & STRATEGY

Key deliverables

Four work products, sequenced from FDA dialogue through substantial equivalence.

1

Q-Sub (Pre-Submission) package

Cover letter · device description · proposed indications · predicates · PCCP summary · specific FDA questions. Validates classification and PCCP scope with FDA before the 510(k).

2

PCCP document

Device description · change categories per module · modification protocols · performance criteria · validation methodology · reporting plan.

3

510(k) submission

Substantial-equivalence argument · software docs · performance / validation data · risk analysis · labeling. Composition driven by composition-manifest.md.

4

Trace matrix per DHF

User Needs ↔ Design Inputs ↔ Architecture ↔ V&V ↔ Risk. Filing Scope column derived from Ct* tags.

AGENTIC APPROACH

Project shape — flat multi-DHF

IEC 62304 § 5 software-system / software-item split, expressed in folders.

```
PDLC_DEMO/
├── CLAUDE.md
├── project.yml
├── project-overview.md
├── tasks/<person>/
├── docs/external/
├── docs/internal/
├── docs/project/strategies/
├── docs/project/dhfs/
│   ├── pca-device/
│   ├── connectivity-adapter/
│   └── cloud-suite/
├── docs/project/submissions/
├── tools/project-console/
└── .claude/
```

- project operating rules (always loaded)
- single source of truth – identity, DHFs, team, registries, secops
- this overview
- per-person task folders; every edit is task-gated
- FDA guidance, ISO/IEC standards, frameworks, literature
- GlobalLogic QMS – Manual + 25 SOPs + 6 WIs + 14 Templates + 9 Forms
- 8 shared strategy docs (regulatory, architecture, dev, testing, ...)
- TOP DHF – PP3500 (lead product)
- TOP DHF – MDDS bridge
- PLATFORM DHF – 7 nested children
 - qsub/ · 510k/ · pccp/ (each with composition-manifest.md)
- local FastAPI console (scaffolded from project-console skill)
- skills/ agents/ hooks/ settings.json

AGENTIC APPROACH

Operating rules baked into the project

Hooks and conventions that make Claude work with regulatory-grade discipline.

Task-first gate

PreToolUse hook denies any Edit/Write/NotebookEdit unless the session has an active task document. Every change is owned.

Strategy / lessons capture

UserPromptSubmit + Stop hooks detect strategic-intent signals and block session end if content wasn't captured.

Docflow entry enforcement

Bash hook refuses direct pandoc / soffice / pdftotext calls so all conversions round-trip through /docflow.

One task, one file

All analysis, drafts, and phase deliverables live in the task's numbered markdown file — never sibling notes.

Session-start secops

project-secops agent audits the session against project.yml allowlists. Results cached with a 7-day TTL.

project.yml as single source of truth

Identity, DHF topology, team, registries, secops allowlists, advisor curation — all in one manifest.

AGENTIC APPROACH

Skills — reusable, expert-grade playbooks

15 project skills installed + 4 builtin format skills. Highest-leverage shown.

SKILL	WHAT IT AUTOMATES
/task	Task lifecycle · enforces the task-first gate · manages capture hooks.
/medtech-docs	Scaffolds docs/ tree · manages DHFs · imports FDA guidance and ISO/IEC standards.
/docflow	Round-trip DOCX / DOC / PDF / XLSX ↔ markdown with image and cross-ref fidelity.
/strategy	Harvests tagged blocks from tasks into 8 shared strategy docs.
/tracker	Builds 510(k) + PCCP submission tracker from strategy + composition manifests.
/trace-matrix	Bidirectional design-controls trace per DHF; emits md + json sidecar.
/advisors	Installs and curates 23 persona advisors; shared between Claude Code and the console.
/project-console	Scaffolds and maintains the local FastAPI console (agents / docs / dashboards / trace).
/lessons	Harvests lessons-learned blocks into the team ledger; promotes mature ones.
/best-practices	Audits project setup against registry checks and each skill's best-practices table.
/secops	Session-start security posture check, attestations, allowlists.
/change-control	(Scaffold.) Bridge to Confluence / Comala and Windchill for Part 11 review & release.
/sync-skills	Bidirectional sync with the shared hitachi skill registry.
/digest	Morning briefing + on-demand append to the project CHANGELOG.md.

AGENTIC APPROACH

Agents — persona advisors

23 total — 11 Core Team Assistants + 8 KOLs + 4 panels. Same definitions feed Claude Code and the console.

11 Core Team Assistants

- Regulatory Affairs · Clinical Affairs
- Systems Engineering · R&D Lead
- V&V Lead · Risk Management
- Human Factors · Quality Engineering
- Cybersecurity · Post-Market · Program Manager
- **+ 2 panels:** Core Team (PM/RA/Clinical/QE/R&D) and Design Review (Sys/R&D/V&V/HFE/Risk/QE).

8 KOLs + 2 panels (project-specific)

- **James E. Paul** PCA pump safety, acute pain — primary PP3500 anchor KOL
- **Kathleen K. Giuliano** IV smart-pump usability, medication safety
- **Priyadarshini Pennathur** Human factors engineering
- **Lisa Gorski** Infusion nursing standards, vascular access
- **Evan Kirkendall / Parth Shah / Sini Kuitunen / Susan Braithwaite** smart-pump safety, alert fatigue, pediatric dosing, insulin protocols
- **+ 2 panels:** round-robin and LLM-moderated PP3500 KOL panels.

AGENTIC APPROACH

Three-tier grounding model

Every advisor has grounding sources layered from most-portable to most-specific.

1

Universal

FDA guidance · ISO/IEC standards · industry frameworks · clinical literature. Identical across every medtech project scaffolded by /medtech-docs.

2

Shape-stable

Paths guaranteed by /medtech-docs — DHF layout, strategies/, submissions/, input-analysis/. The advisor knows where to look, regardless of project.

3

Project-specific overlay

project.yml →
advisors.overlays.<name>.add / .exclude narrows the file set to what this program actually cares about (e.g. PCA regs, not hip-surgery regs).

QUALITY & PROCESS

Guardrails that make quality automatic

Skills, hooks, and rules turn written conventions into executed ones.

Skills — reusable playbooks

Named, one-command actions that every team member runs the same way.
Remove inconsistency at the source.
/task, /medtech-docs, /docflow,
/tracker, /trace-matrix, /best-practices.

Hooks — automatic tripwires

Small automations fired at a specific moment (session start, before an edit, when work ends) that allow, block, or add context. The system refuses to let you drift.

Rules — conventions the tool obeys

Short conventions in
CLAUDE.md / .claude/rules/ that Claude reads before acting. One task one file ·
READMEs have Conventions + Changelog
· never fabricate standards or clinical content · demo banner everywhere.

QUALITY & PROCESS

Concrete automations, in plain language

The hooks running today. What each one refuses to let you skip.

Task-first gate

No file edit without an active, owned task.

Session env

Exports CLAUDE_SESSION_ID — the task gate keys its state on this.

Docflow entry

Raw pandoc / soffice / pdftotext Bash calls refused — must go through /docflow.

Session security check

16 automated checks (identity, access, supply chain). High-severity failures block work.

Strategy / lessons capture

If a prompt shows strategic intent but the task never captured a STRATEGY block, Stop is blocked.

Session cleanup

Per-session state files purged on SessionEnd.

HUMANS IN CHARGE

Agents advise. Humans decide.

Cross-functional expertise on tap — with the decision-maker of record always a named human.

Advisors do

- Brief the human with a structured, grounded analysis.
- Cite sources for every claim.
- Surface counterpoints automatically.
- Refuse to fabricate — say "source doesn't support this."
- Leave a paper trail of what they read and said.

Humans do

- Review the analysis · ask follow-ups.
- Choose the answer (agent may have given three).
- Write the decision into the task document.
- Sign off in the QMS of record — Part 11 signature by a named human.
- No AI-generated content enters a controlled artifact without human review.

What the regulator sees

- A human-signed decision.
- In a task document owned by a named person.
- Backed by a trace matrix rebuilt from source.
- Supported by structured analysis with cited sources.
- Vetted by multiple specialist perspectives.
- Produced in a toolchain whose compliance properties are enforced by hooks, not hope.

HUMANS IN CHARGE

Where the handoff happens

The AI never occupies step 3 or step 6. That is the design, not a limitation.

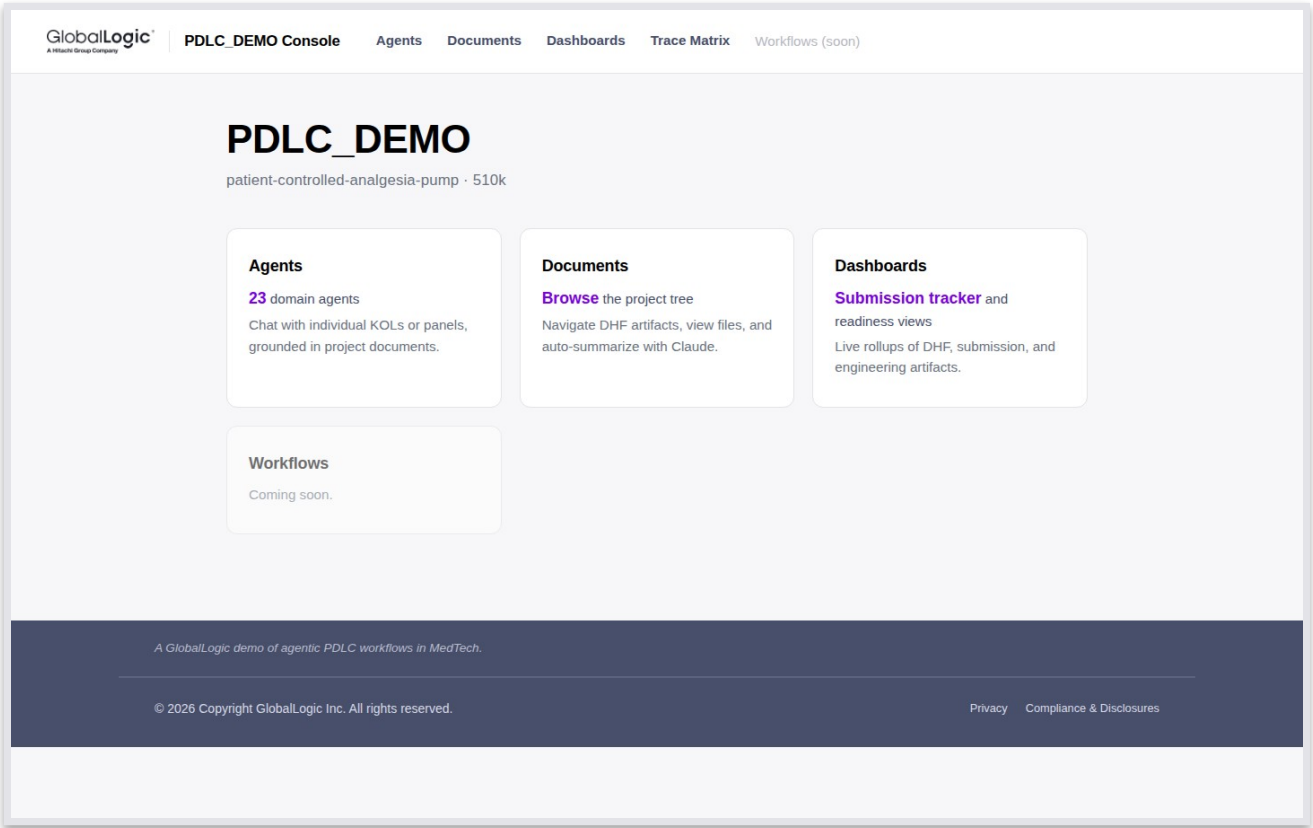
#	STEP	WHO	WHAT
1	Draft analysis	Agent	Reads grounded sources, drafts structured answer with citations + counterpoints.
2	Review & redirect	Human	Reviews the analysis, asks follow-ups, rejects or redirects.
3	Decision	Human	Chooses the path forward and writes it into the task document.
4	Implementation	Human + agent	Human directs; agent drafts; task-first gate ensures every edit is owned and traced.
5	Verification	Human (or panel)	Independent panel agent reviews before the change is finalized.
6	Sign-off in QMS of record	Human only	Controlled artifact promoted; signatures applied by named humans in the system of record.

THE PROJECT CONSOLE

Landing page

Top-level tiles for Agents, Documents, Dashboards, and Trace Matrix — branded with the GlobalLogic theme.

<http://127.0.0.1:8765/>



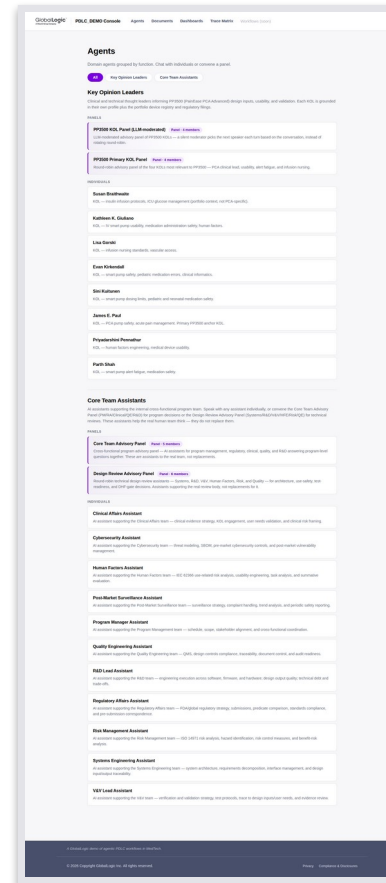
Start / restart: `/project-console start` or `./tools/project-console/run.sh`

THE PROJECT CONSOLE

Agents — 23 domain agents

11 Core Team Assistants + 8 KOLs + 4 panels. Curated via `project.yml` → `advisors.enabled`.

<http://127.0.0.1:8765/agents>

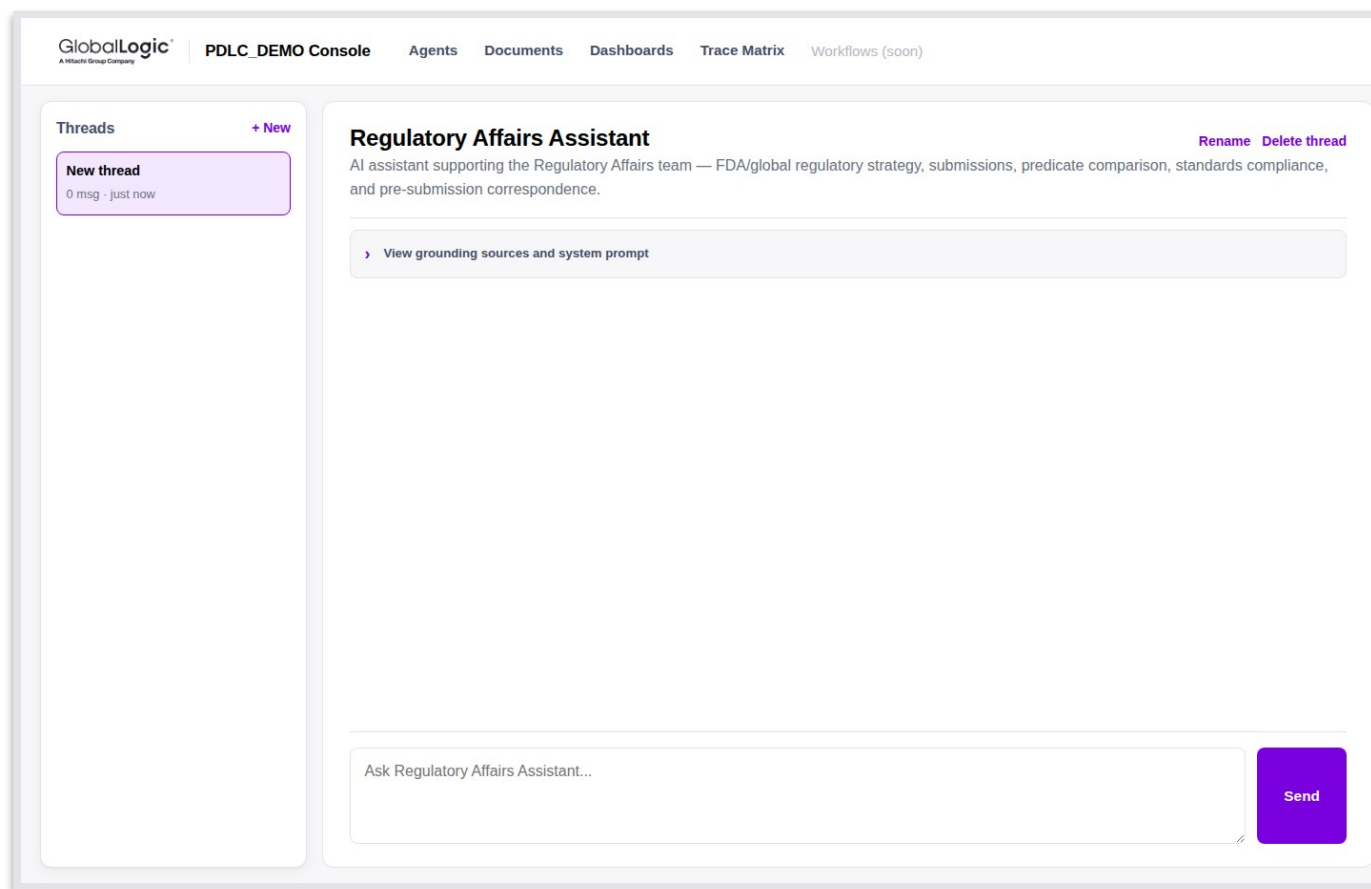


THE PROJECT CONSOLE

A single agent chat

Threaded, renameable, with a collapsible grounding-sources + system-prompt panel.

<http://127.0.0.1:8765/agents/regulatory-affairs>

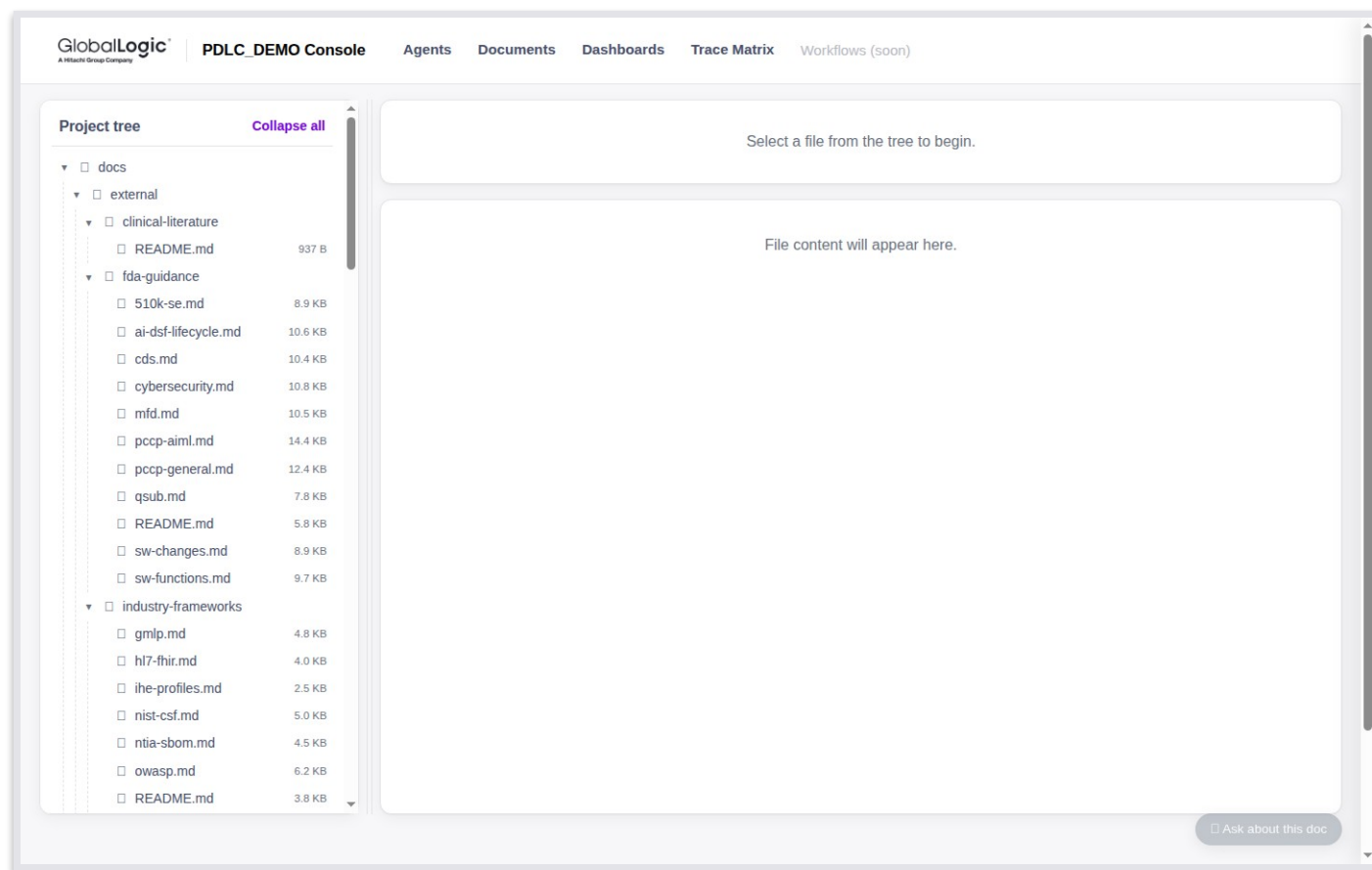


THE PROJECT CONSOLE

Documents explorer

Full project tree · inline markdown rendering · unified Assistant drawer (ben/024) grounded on the open doc.

<http://127.0.0.1:8765/documents>

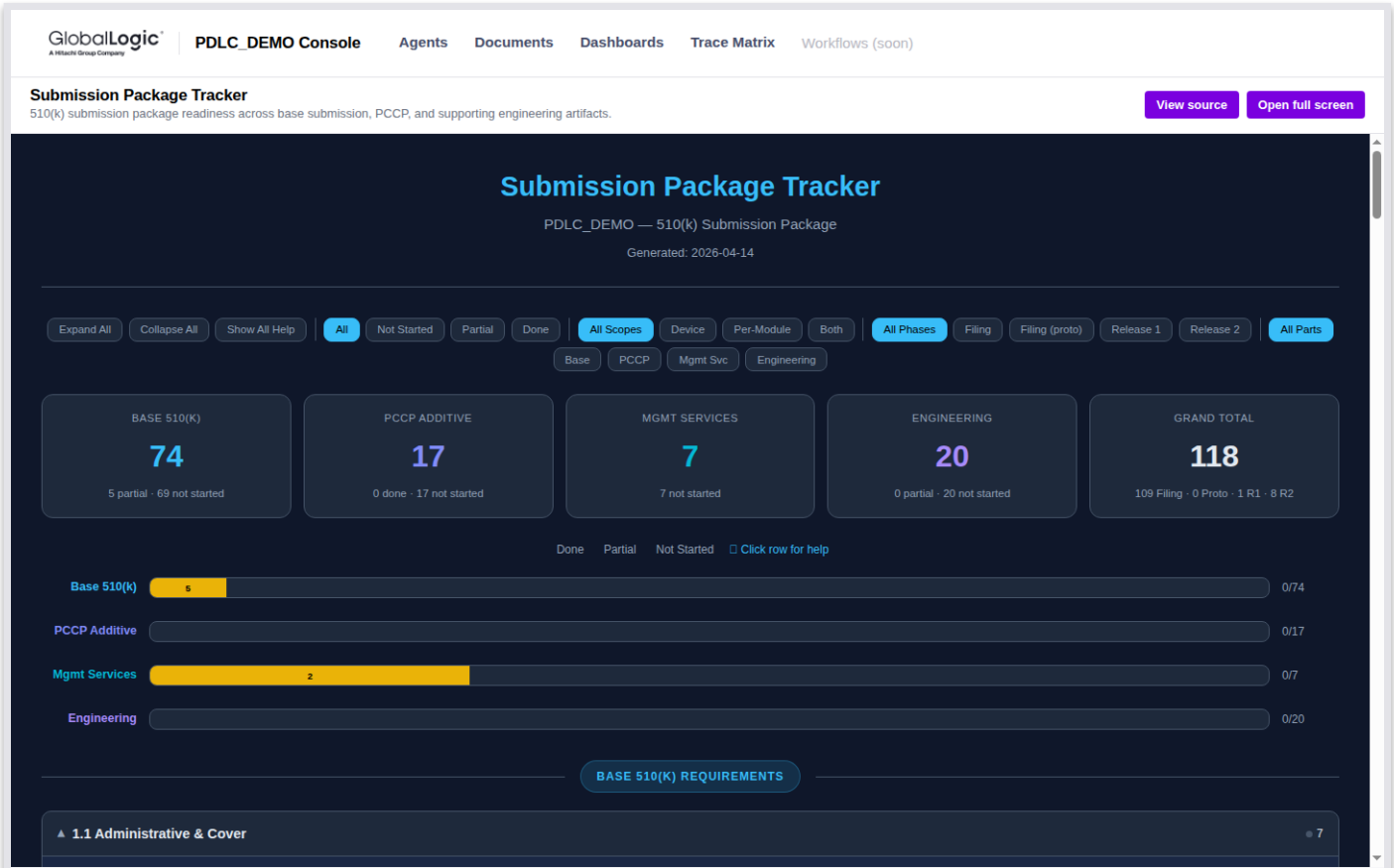


THE PROJECT CONSOLE

Dashboards — submission tracker

Glob-discovered HTML dashboards rendered inline. Today: Submission Package Tracker (built by /tracker).

<http://127.0.0.1:8765/dashboards/submission-tracker>



THE PROJECT CONSOLE

Trace matrix — pca-device detail

UN 22 · DI 34 · Arch 7 · V&V 3. Ct* filters, orphans-only toggle, SE Assistant drawer.

<http://127.0.0.1:8765/trace-matrix/pca-device>

GlobalLogic[®]
A Hitachi Group Company

PDLC_DEMO ConsoleAgentsDocumentsDashboardsTrace MatrixWorkflows (soon)

Trace Matrix — pca-device skill v1.1 - 2026-04-15T04:56:15Z

SE AssistantRebuild

User Needs 22Design Inputs 3439 ORPHANArchitecture 77 ORPHANEDGES UNKNOWNVerification & Validation 3Risk 0NO_ITEMS_PARSED

CRITICALITYAllCISCIFCICCPSuppUntaggedAll rowsOrphans onlyExpand allCollapse all

ID	REQUIREMENT	CRIT	GROUP	.. FORWARD	.. REVERSE
DI-001	The device shall deliver programmed continuous flow rates from 0.01 to 25 mL/hr with an accuracy of ±5% (or ±0.5 mL/hr, whichever is greater) across the full operating range.	CIS	G1	VER-PP3500-BT-005	UN-001UN-002
DI-002	The device shall deliver patient-activated bolus doses from 0.01 to 20 mL with a delivered-volume accuracy of ±5% across the dose range.	CIS	G1	—	UN-001
DI-003	The device shall support a programmable lockout interval from 1 to 99 minutes between consecutive PCA bolus requests.	CIF	G1	VER-PP3500-SW-002	UN-003
DI-004	The device shall enforce programmable 1-hour and 4-hour cumulative dose limits and shall block any bolus that would exceed the limit.	CIS	G1	VER-PP3500-SW-002	UN-003UN-004
DI-005	The device shall enforce drug-library hard maximum limits such that no programming path — manual entry, barcode-loaded order, or clinician override — can program a value exceeding the active library's hard cap.	CIS	G2	VER-PP3500-SW-006	UN-004UN-009
DI-015	The device shall maintain an on-board drug library of at least 200 medications with editable hard and soft limits, and shall display the active library version on the home screen and at therapy start.	CIF	G2	—	UN-009
DI-016	The device shall include an integrated 1D/2D barcode scanner capable of reading medication container barcodes (NDC, GS1) and matching them against the active drug library before therapy start.	CIF	G2	—	UN-008
DI-007	The device shall detect downstream occlusions in the range 3 to 15 psi and annunciate a high-priority occlusion alarm within the response time required by IEC 60601-2-24.	CIS	G3	—	UN-006
DI-008	The device shall detect single air-in-line bubbles ≥50 µL and cumulative air ≥1 mL over a 15-minute window and shall annunciate a high-priority air-in-line alarm.	CIS	G3	—	UN-006
DI-009	The device shall annunciate a high-priority over-infusion alarm and halt delivery if the measured delivered volume deviates from the programmed volume by more than 10% over any rolling 1-hour window.	CIS	G3	—	UN-001UN-006
DI-019	The device shall provide audible alarms compliant with IEC 60601-1-8 for high-, medium-, and low-priority alarm categories, with sound pressure level adjustable from 45 to 80 dB(A) at 1 m.	CIF	G3	—	UN-010

Headline gap — 30 Design-Input orphans — surfaced exactly as the trace matrix is designed to.

APPENDIX

Quick links

Everything referenced in this deck.

RESOURCE	LOCATION
Project operating rules	CLAUDE.md
Project manifest	project.yml
Companion markdown	project-overview.md
Regulatory strategy	docs/project/strategies/regulatory-strategy.md
Architecture strategy	docs/project/strategies/architecture-strategy.md
PCA device DHF	docs/project/dhfs/pca-device/
Cloud Suite DHF	docs/project/dhfs/cloud-suite/
510(k) composition manifest	docs/project/submissions/510k/composition-manifest.md
Submission tracker (md)	docs/project/submissions/submission-tracker.md
Console landing	http://127.0.0.1:8765/
Console agents	http://127.0.0.1:8765/agents
Console documents	http://127.0.0.1:8765/documents
Console dashboards	http://127.0.0.1:8765/dashboards
Console trace matrix	http://127.0.0.1:8765/trace-matrix

Thank you

Questions & discussion